

Calibration Reference -- Cameras, Markers & Coordinate System

Cross-referenced from , , and .

Coordinate System

Axis	Direction	Notes
Origin (0,0,0)	Back right corner of Panel Unit 0, at floor level	Where the rightmost panel meets the storefront floor
X	Negative = left (toward Unit 3)	Panels span X = 0 to X = ?300
Y	Positive = up	Floor = 0, street is below
Z	Positive = forward (away from panels, into tracking zone)	Cameras face this direction

Reference Y Levels

Level	Y (cm)	Description
Storefront floor	0	Origin reference plane
Camera mount	?15	15 cm below floor (see discrepancy note below)
Street / sidewalk	?66	Where people walk, 66 cm below floor

Camera Positions

Both cameras are mounted at the front edge of the active tracking zone (Z = 78), looking outward into the street.

	Camera 1 (RIGHT)	Camera 2 (LEFT)
Position	(?30, ?15, 78)	(?270, ?15, 78)
Aligned with	Unit 0 center	Unit 3 center
Color in viz	Red	Blue
IP	10.42.0.75	10.42.0.172
Model	Reolink RLC-520A	Reolink RLC-520A
FOV	80 deg H x 48 deg V	80 deg H x 48 deg V
Looks toward	(?150, ?66, 180) -- center of active zone	(?150, ?66, 180) -- center of active zone

Camera Angles (Euler XYZ, degrees)

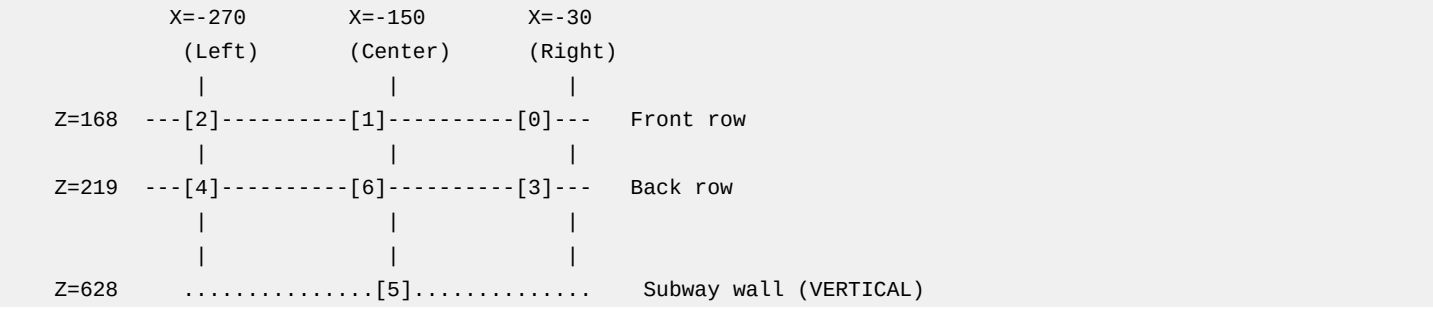
	Pitch (tilt down)	Yaw (horizontal rotation)	Roll
Camera 1	22 deg down	?25 deg (angled left toward center)	0 deg
Camera 2	22 deg down	+25 deg (angled right toward center)	0 deg

Both cameras aim inward to create an overlap zone around X = ?100 to X = ?200 for stereo matching. All files now agree on pitch = 22 deg and yaw = +/-25 deg.

Calibration Markers

7 ArUco markers (DICT_4X4_50), each 15 cm x 15 cm.

Marker Map (bird's eye, looking down from above)



Marker Positions

ID	Position (X, Y, Z)	Orientation	Description	Visible to
0	(?30, ?66, 168)	Flat on ground	Right front	Camera 1 only
1	(?150, ?66, 168)	Flat on ground	Center front (SHARED)	Both cameras
2	(?270, ?66, 168)	Flat on ground	Left front	Camera 2 only
3	(?30, ?66, 219)	Flat on ground	Right back	Camera 1 only
4	(?270, ?66, 219)	Flat on ground	Left back	Camera 2 only
5	(?150, ?15, 628)	Vertical on wall	Subway wall (~550 cm from cameras)	Both cameras
6	(?150, ?66, 219)	Flat on ground	Center back (SHARED)	Both cameras

- Front row is 90 cm forward from the tracking zone edge ($Z = 78 + 90 = 168$)
- Back row is 51 cm behind front row ($Z = 168 + 51 = 219$)
- Marker 5 is on the subway wall, ~550 cm from camera line ($Z = 78 + 550 = 628$)

Per-Camera Visibility

Camera 1 (Right)	Camera 2 (Left)
0, 1, 3, 5, 6	1, 2, 4, 5, 6

Shared markers: 1, 5, 6 -- these provide the common reference frame between cameras. Minimum 3 markers required per camera for solvePnP.

Tracking Zones

Defined in (the controller owns zone classification, not the tracker):

Active Zone (people engaging with the installation)

Dimension	Value	Computed bounds
Width	260 cm	X: ?280 to ?20
Depth	205 cm	Z: 78 to 283
Height	300 cm	Y: ?66 to 234
Center X	?150	Centered on panels

Passive Zone (sidewalk passersby)

Dimension	Value	Computed bounds
Width	400 cm	X: ?350 to 50
Depth	270 cm	Z: 283 to 553
Height	300 cm	Y: ?66 to 234

Discrepancies -- RESOLVED

All four discrepancies identified during the initial cross-file audit have been fixed:

#	Issue	Fix Applied
1	was ?16 in both files	Changed to ?15 in and
2	Marker 5 Z was 578 in and	Changed to 628 (matching the fallback and physical measurem
3	Tracking zone bounds in were from an old coordinate cor	Updated to match : active X [?280,?20], Z [78,283]; passive
4	Camera pitch was 10 deg in	Changed to 22 deg to match

All three files (, ,) are now in agreement on all coordinate values.